18-813 / 19-746: Foundations of Responsible AI

Assignment 1: Syllabus and Fairness

Fall 2026

Released: Monday, August 24, 2026

Due: Wednesday, September 2, 2026, 12 PM ET

How to submit. Upload your submission to Gradescope by 12 PM ET on the due date. Read the Syllabus section on Gradescope for details. Submissions may receive a score of 0 for any questions that do not follow the instructions on the Syllabus.

Collaboration. You may (and are encouraged to) collaborate with classmates, but you must write up your own solutions. Identical solutions are not permitted. Please read the Syllabus for the full collaboration policy.

AI usage. You may use AI to help you understand the material, brainstorm, format, or proofread. You may not use AI as a substitute for learning the material or generating substantive contributions. Please read the Syllabus for the AI usage policy. As mentioned in the Syllabus, we may ask you to explain any submitted work and use your response to score your submission.

Late policy. Assignments are due at 12 PM ET. You have 3 late days for the semester, which apply to assignments only. After late days are used, 10% of the assignment total is deducted per day. Please see the Syllabus for details.

Question #1: Course logistics [4 points]

Answer the following based on the course syllabus. Short answers are sufficient.

(a) Where should you post general questions about assignments, homework, or project logistics? One word is sufficient.

(b) Where will announcements, course notes, assignments, and readings be posted? One word is sufficient.

(c) What happens if you repeatedly mislabel where each question appears in your Gradescope upload?

(d) What is the default grade if a grader cannot read your submission?

Question #2: Grading, late days, and make-ups [5 points]

(a) At what time of day are assignments due?

(b) What happens after two unsuccessful re-grade requests?

(c) How many assignment late days do you have for the semester?

(d) After late days are used up, what is the penalty per day?

(e) If you anticipate missing a quiz or project presentation, how far in advance must you alert the instructor and TAs?

Question #3: Collaboration, AI, and other policies [5 points]

(a) What is the grade penalty if plagiarism or cheating is discovered?

(b) Are you allowed to use AI to help you understand the material, brainstorm, format, or proofread?

(c) Are you allowed to use AI to generate substantive contributions?

(d) True or false: The teaching staff may ask you to explain how you solved a problem or reached a conclusion, and may use your response in grading.

(e) May you record or transcribe lectures?

Question #4: Probability and statistics [12 points]

Below, short derivations or one-line answers are sufficient unless a proof/calculation is requested.

(a) Fill in the blanks. If A and B are independent,

$$\begin{aligned}\mathbb{P}(A \cap B) &= \underline{\hspace{6cm}},\\ \mathbb{P}(A, B) &= \underline{\hspace{6cm}},\\ \mathbb{P}(A \mid B) &= \underline{\hspace{6cm}},\\ \mathbb{P}(B \mid A) &= \underline{\hspace{6cm}}.\end{aligned}$$

(b) Use Bayes' rule to express $\mathbb{P}(B \mid A)$.

(c) Use marginalization to express $p(x)$ in terms of the joint $p(x, y)$ if Y is discrete.

(d) Use marginalization to express $p(x)$ in terms of the joint $p(x, y)$ if Y is continuous.

(e) Suppose Z is a random variable taking values $z_1, \ldots, z_k \in \mathcal{Z}$. Write $\mathbb{E}[Z]$.

(f) A hospital records recovery rates with and without a treatment, by patient group:

	Treatment	No treatment
Group 0	$8/10$ recovered	$70/90$ recovered
Group 1	$15/90$ recovered	$3/20$ recovered

Compute the recovery rates with and without treatment (i) within Group 0, (ii) within Group 1, and (iii) overall. You should end up with six numbers. What do you observe? What is this phenomenon called, and why does it matter when you only look at aggregate statistics?

Question #5: Optimization [8 points]

Consider a function f_θ parameterized by θ (for example, a predictor with weights θ) that generates a prediction for inputs/features $x \in \mathcal{X}$.

(a) Write an objective that *minimizes expected loss* of f_θ. Use any notation from the lecture that you feel is relevant.

$$\min_{\theta} \underline{\hspace{7cm}}.$$

(b) Suppose $f_\theta \in [0, 1]$ is higher for better outcomes. Write an objective that *maximizes the worst-case outcome* of f_θ over a collection of inputs $x_i, \ldots, x_k$:

$$\max_\theta \underline{\hspace{6cm}}.$$

(c) Let D be a training dataset, and consider the constrained problem

$$\min_\theta \text{Loss}(f_\theta, D) \qquad \text{s.t.} \qquad g(\theta) \leq \delta.$$

Write a Lagrangian for this problem, using λ as the Lagrange multiplier.

(d) What happens, intuitively, if λ increases? If it decreases?

Question #6: Machine learning [8 points]

Consider a standard supervised learning setup: features X, label Y, model f_θ, and a prediction $\hat{Y}$ obtained from $f_\theta(X)$ (often by thresholding).

(a) Suppose we have samples $\{(x_i, y_i)\}_{i=1}^n$. Write the empirical loss of f_θ over these samples using any notation from the lecture that you feel is relevant.

(b) In one sentence, describe overfitting, and give one typical symptom in terms of the training and validation error curves.

(c) A binary classifier outputs $\hat{Y} \in \{0, 1\}$ and the true label is $Y \in \{0, 1\}$. Define true positive rate (TPR), true negative rate (TNR), false positive rate (FPR), and false negative rate (FNR) as probabilities.

(d) These four rates are redundant. Express FNR in terms of TPR, and FPR in terms of TNR.

Question #7: Responsible AI [6 points]

(a) The lecture highlights two factors of AI that make it important to consider how it affects society and individuals. What are they? Briefly define each.

(b) Using the hiring example from lecture, why does the first factor (autonomy) matter?

(c) Using the hiring example from lecture, why does the second factor (automation) matter?

Question #8: How unfairness arises [12 points]

(a) In hiring, the model is trained on pairs (X, Y), where X is the applicant's features (e.g., a resume) and Y is the label used as the training outcome. We often cannot observe the outcome we actually care about (e.g., whether the person would succeed on the job), so we train on a *proxy* Y instead (e.g., whether they were hired, or their later salary). What is the fairness problem with this? Your answer must show sufficient understanding of the specific issue.

(b) Suppose the covariates X are not descriptive enough. In other words, there are 3 relevant features $X_1 \in \{0, 1\}$, $X_2 \in [0, 1]$, and $X_3 \in [0, 1]$ but $X = (X_1, X_2)$. Why does this result in a fairness problem? You may assume an applicant for which $X_1 = 0$ is only qualified if $X_3 < X_2$, but an applicant for which $X_1 = 1$ is only qualified if $X_3 > X_2$. You can express your answer in words, but be clear about its relation to X_1, X_2, X_3.

(c) Suppose the model class is underparameterized (not expressive enough). What is the fairness problem?

(d) How can overfitting contribute to unfairness?

(e) Why does the *selective labels* problem in a hiring example where the training data contains examples (X, Y) of whether applicants X succeed in the job as denoted by Y result in fairness issues?

(f) The model is trained to minimize average loss, so every sample is weighted equally. Why is that a fairness problem when the data are imbalanced?

Question #9: Fairness definitions [8 points]

Let $Y \in \{0,1\}$ be the true outcome, $\hat{Y} \in \{0,1\}$ a binary prediction, and $A \in \{0,1\}$ a sensitive group attribute. Write each group-fairness definition in probability notation (e.g., $\mathbb{P}(\cdot) = \mathbb{P}(\cdot)$), then give a one-line interpretation. (Note that, in practice, fairness only requires approximate equality, not exact equality.)

(a) Demographic parity.

(b) Equal opportunity.

(c) Equalized odds.

(d) Predictive parity.

Question #10: Individual and intersectional fairness [12 points]

Group fairness (Question 9) partitions people by a sensitive attribute $A \in \{0,1\}$ and requires the two groups $\{A=0\}$ and $\{A=1\}$ to be treated similarly in some average sense (equal selection rates, equal TPR, etc.). Individual fairness is a different idea: similar people should be treated similarly. While these two are sometimes compatible, they don't need to be.

(a) Why does group fairness sometimes prevent individual fairness?

Now suppose instead of *one* sensitive attribute, we have *two* sensitive attributes, $A \in \{0,1\}$ and $B \in \{0,1\}$. Suppose a hiring model produces the following selection counts (i.e., selection rates given $\hat{Y}$):

	$B=0$	$B=1$
$A=0$	$40/50$ selected	$10/50$ selected
$A=1$	$10/50$ selected	$40/50$ selected

(b) Compute the selection rates (as a percentage) for $A=0$ vs. $A=1$. Does demographic parity hold for A?

(c) Compute the selection rates (as a percentage) for $B=0$ vs. $B=1$. Does demographic parity hold for B?

(d) Compute the four intersectional selection rates $\mathbb{P}(\hat{Y}=1 \mid A=a, B=b)$ for all pairs $(a,b) \in \{0,1\}^2$.

(e) Does demographic parity hold across all subgroups (A, B), i.e., for $(A, B) = (0,0)$, $(0,1)$, $(1,0)$, and $(1,1)$?

(f) Thus, intersectional fairness is a form of "group fairness" on a smaller group. The lecture says you can require group fairness properties on increasingly smaller groups, and in the limit a group is a single x, which would actually guarantee individual fairness. Thus, intersection fairness sits between broad group fairness notions and individual fairness. However, there is a practical cost related to performance. What is it?

Question #11: Fairness methods [8 points]

(a) What is the optimization / objective-function approach to fairness? Write an objective (or constrained problem) over model parameters θ using notation from lecture.

(b) Suppose we want group fairness with respect to A. Why does simply removing A from the input features generally *not* make $\hat{Y}$ independent of A?

(c) What does reweighting the training data with respect to the covariates X do, and what problem is it intended to address?

(d) In adversarial debiasing, what is the predictor trying to do, and what is the adversary trying to do?

Question #12: An impossibility result [12 points]

In Question 9 you defined three group-fairness notions: demographic parity, equal opportunity (equal true positive rates), and predictive parity. Lecture also noted that some fairness notions are incompatible: they cannot hold exactly at the same time (though approximate versions sometimes can). In this question you will show one such impossibility.

As before, let $Y \in \{0,1\}$ be the true outcome, $\hat{Y} \in \{0,1\}$ a binary prediction, and $A \in \{0,1\}$ a sensitive group attribute. Throughout, assume all conditioning events have positive probability (so the conditionals below are well-defined). Recall that

$$\mathrm{TPR}_a \;=\; \mathbb{P}(\hat{Y}{=}1 \mid Y{=}1, A{=}a).$$

(a) Rewrite $\mathbb{P}(Y{=}1 \mid \hat{Y}{=}1, A{=}1)$ using Bayes' rule.

(b) Rewrite $\mathbb{P}(Y{=}1 \mid \hat{Y}{=}1, A{=}0)$ using Bayes' rule.

(c) If predictive parity must hold, what is the relationship between the two expressions in (a) and (b)?

(d) If demographic parity also holds, what happens to the equation in (c)? That is, what must hold?

(e) You should be left with an equation involving the TPRs of groups $A{=}0$ and $A{=}1$. What must be true in order for the TPRs to be equal?

(f) Part (e) implies that demographic parity (DP), predictive parity (PP), and equal opportunity (EO) cannot hold simultaneously unless a nearly impossible condition holds. Thus, a designer must choose between them. Which would you choose? Why?

Question #13: Required statements

Complete both statements. You will not be penalized simply for collaborating or for using AI.

(a) Collaboration statement: *I discussed this assignment with: [names, or "no one"]. I used the following resources: [list, or "course materials only"].*

(b) AI-usage statement: *I used AI tools as follows: [describe, or "I did not use AI tools"].*